

# HAT-P-36b

R-0847

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_HATP-36\_240408 · created 2026-07-28T20:11:55+00:00

## BENCHMARK: MARGINAL

### Results

|                                                   |                                |
|---------------------------------------------------|--------------------------------|
| Mid-Transit Time (Tmid)                           | 2460408.6785 +/- 0.004 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.103 +/- 0.011                |
| Transit depth (Rp/Rs)^2                           | 1.06 +/- 0.24 [%]              |
| Orbital Inclination (inc)                         | 81.93 +/- 2.7                  |
| Airmass coefficient 1 (a1)                        | 1.0726 +/- 0.0071              |
| Airmass coefficient 2 (a2)                        | -0.066 +/- 0.017               |
| Scatter in the residuals of the lightcurve fit is | 0.67 %                         |
| Best Comparison Star                              | #2 - [93, 180]                 |
| Optimal Aperture                                  | 4.11                           |
| Optimal Annulus                                   | 13.45                          |
| Transit Duration (day)                            | 0.076 +/- 0.016                |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                                      |                                                       |
|--------------------------------------|-------------------------------------------------------|
| Rp/R $\blacksquare$ fit vs published | 0.103 $\pm$ 0.011 vs 0.126 (deviation 2.07 $\sigma$ ) |
| Duration fit vs published            | 0.076 d vs 0.0925 d (deviation 1.02 $\sigma$ )        |
| Mid-time O-C                         | 11.6 min                                              |
| Depth SNR                            | 9.3                                                   |
| Residual scatter                     | 0.67 %                                                |

### Light curve

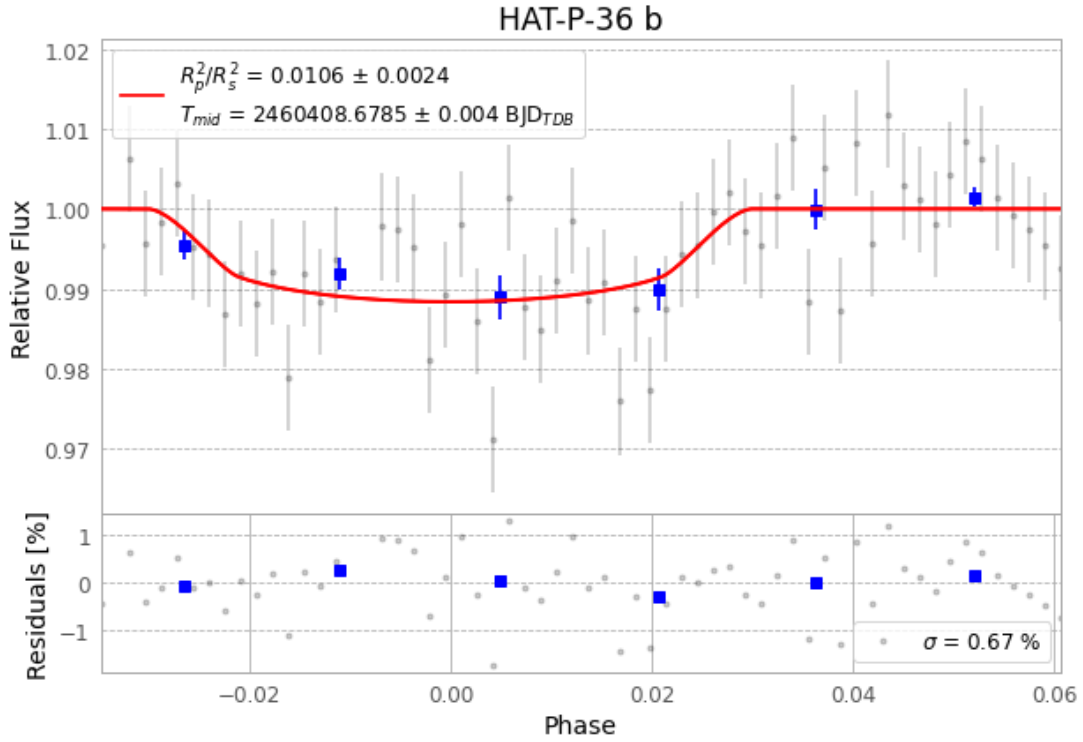


Figure 1 — FinalLightCurve\_HAT-P-36 b\_2024-04-07.png (SHA-256 3c2229e458f52e11...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [365, 244], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 f2c8cc00a0508636...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ead60b9

## Data provenance

61 FITS frames (40 MB), HATP-36240408030346.FITS → HATP-36240408060615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-240408.log (0bbcb7250805...), FinalLightCurve\_HAT-P-36 b\_2024-04-07.png (3c2229e458f5...), FinalLightCurve\_HAT-P-36 b\_2024-04-07.csv (dfe69f555770...)

## Compute resources

Wall time 115.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 20:11Z.